

# NHIs in Cybersecurity

Identity is the new perimeter



# Agenda

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1

## Evolution of Cybersecurity

Network → Endpoint → Cloud → Identity

2

## Identity as Control Layer

Why identity is now the new perimeter

3

## Market & Subsegments

Infrastructure · SaaS · Access · Behavior

4

## Fragmentation Analysis

Why no unified platform exists yet

5

## Investment Thesis

Platform opportunity & exit landscape

# The Evolution of Cybersecurity



1990s–2000s

## Network Security

Firewalls · IDS/IPS  
Perimeter defense



2005–2015

## Endpoint Security

AV · EDR · XDR  
Device-level control



2015–2020

## Cloud Security

CASB · CSPM  
Workload protection



2020–Now

## Identity Security

IAM · NHI · AI Agents  
Control layer

# What Is Identity Today?

144:1

Machine-to-Human  
Identity Ratio

+44% YoY Growth in NHIs

Source: Entro Labs NHI & Secrets Risk  
Report H1 2025

Most enterprise activity is now non-human.



Human Users

Workforce & consumer accounts



API Keys & Tokens

Machine credentials



Service Accounts

System-to-system auth



AI Agents

Autonomous software actors



Workload Identities

Cloud roles & compute



# The Identity Security Stack



## Infrastructure Identity

◆ Deepest control over credentials, secrets & system access

⚠ Weak SaaS & post-access visibility



## SaaS / Application Identity

◆ Visibility into app connections, OAuth grants & delegated permissions

⚠ Limited enforcement & little infrastructure control



## Access Control

◆ Dynamic permissions through JIT & least privilege

⚠ Depends on upstream identity systems; can create friction

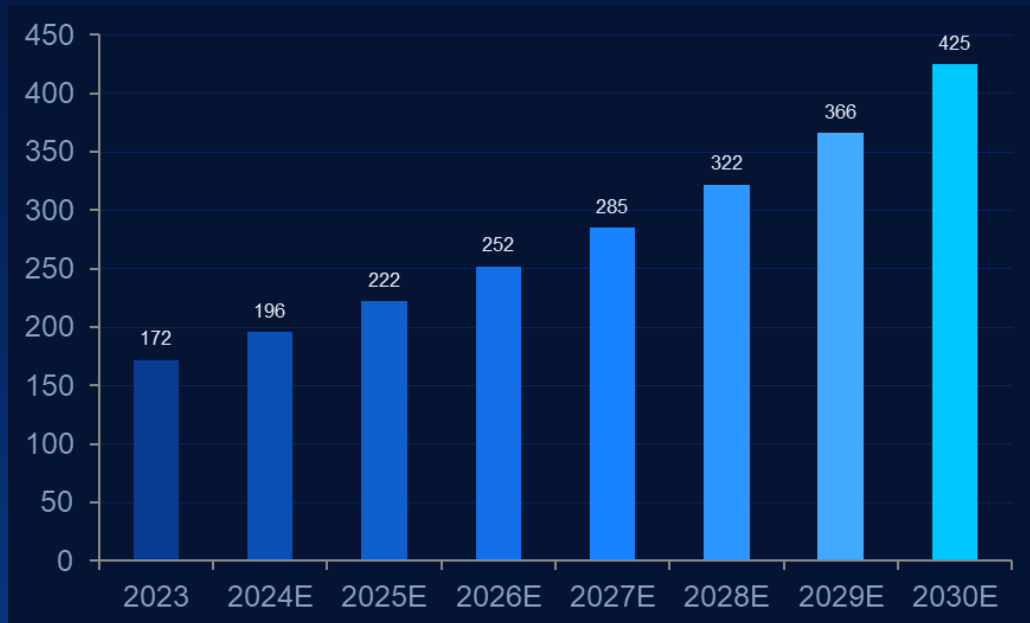


## Behavior / Monitoring

◆ Detects anomalies, misuse & privilege abuse

⚠ Reactive, overlaps with SIEM/UEBA

# Global Cybersecurity Market



**\$172B**

2023 Market Size

**\$425B**

2030 Projection

**~13.8%**

CAGR (2023–2030)

**\$96B**

M&A Value 2025

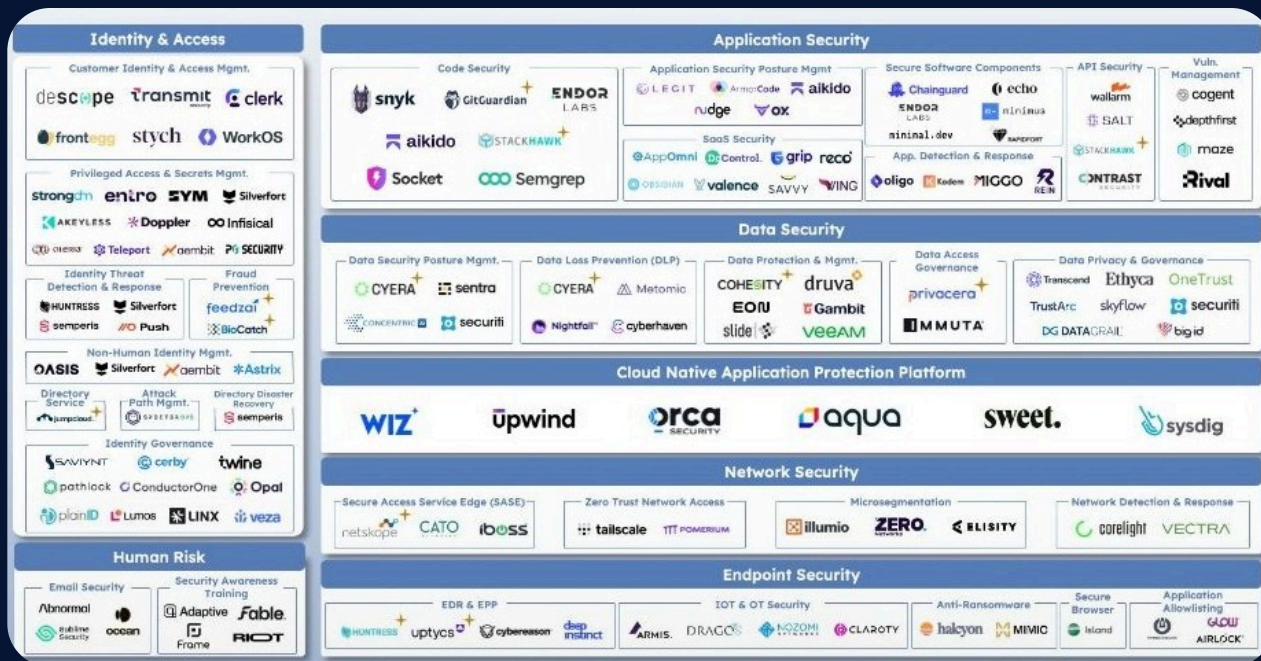
**\$20.7B**

Financing 2025

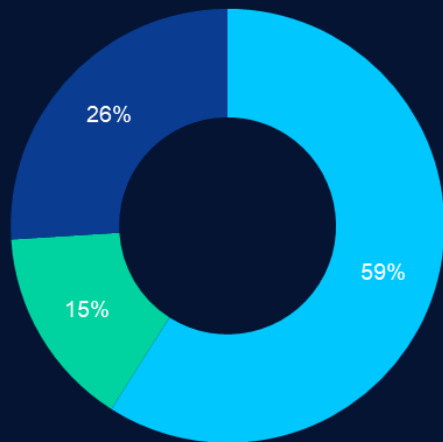
**400**

M&A Deals 2025

# Cybersecurity Market Map



# What's Driving Identity Growth



■ Cloud-Native SaaS ■ AI Security ■ Other Cyber



## Cloud Adoption

No perimeter, identities span multiple clouds & SaaS platforms



## AI & Automation

NHIs grew 44% YoY; AI agents multiply machine identity sprawl



## SaaS Proliferation

59% of M&A volume; 22% CAGR in SaaS cyber since 2019



## Capital Flow

144 AI security financings; \$96B M&A at all-time highs

# Regulatory Tailwinds



## Established Compliance

- ✓ SOC 2 / ISO 27001: user access audits
- ✓ HIPAA / PCI-DSS: privileged access controls
- ✓ GDPR: data access governance
- ✓ FedRAMP / CMMC: federal identity standards

## Emerging / In Progress

- ✓ NHI governance frameworks (early stage)
- ✓ AI agent accountability regulation
- ✓ SEC cyber disclosure rules
- ✓ Zero Trust executive mandates (US Gov)

## Why This Matters for Investors

Regulation converts identity tools from 'nice to have' into durable systems of record, creating sticky, recurring revenue and long replacement cycles.

# Infrastructure Identity: Overview



## API Keys & Tokens

Machine credentials for service auth



## Cloud IAM Roles

AWS/Azure/GCP workload identities



## Service Accounts

Persistent system-to-system identities



## SSH Keys & Certificates

Infrastructure-level cryptographic trust



## Secrets & Vaults

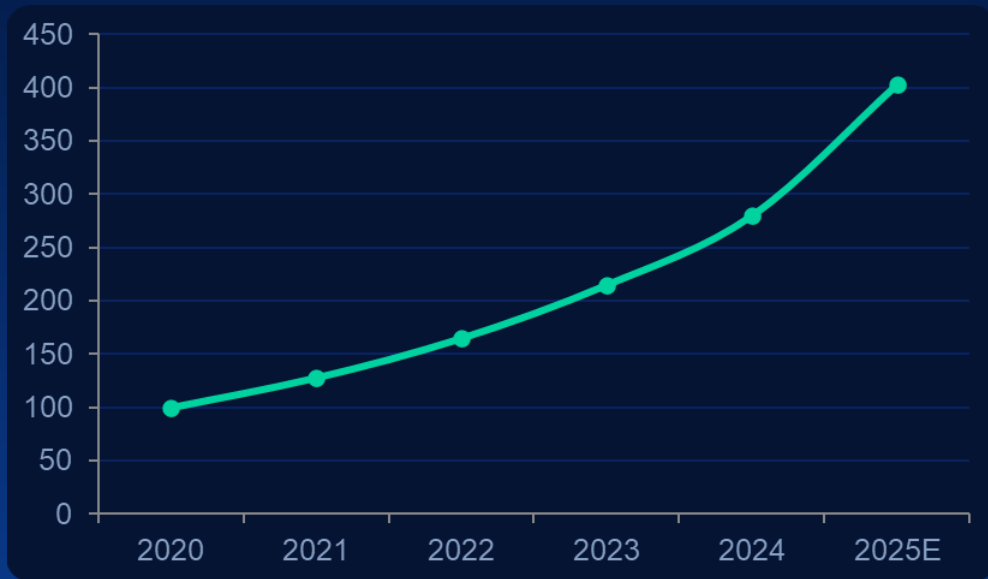
Centralized credential management



## Workload Identity

Short-lived, just-in-time credentials

# Infrastructure Identity: Market



44%

YoY NHI Growth

144:1

Machine-to-Human Ratio

Multi

Cloud Complexity

CI/CD

DevOps Acceleration



# Infrastructure Identity: Limitations



Weak visibility into SaaS integrations & delegated permissions



Limited app-level OAuth & third-party relationship control



Minimal post-access behavior monitoring



Hard multi-cloud identity discovery & correlation



Legacy IAM architectures not built for NHI scale or sprawl



# Infrastructure Identity: Investment Activity

\$25B

2025

Palo Alto Networks

Acquired: CyberArk (Machine Identity)

\$32B

2025

Google

Acquired: Wiz

\$26.4B

2025

IAM Category Total

Acquired: Disclosed M&A Value



# Infrastructure Identity — Thesis

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Closest to credentials, the ultimate enforcement layer



Strongest control point available today



Best starting point for platform expansion



Highest strategic M&A value (\$25B–\$32B exits)



Not a complete platform alone, lacks SaaS, behavioral layers

# SaaS Identity: Overview



## OAuth Permissions

App-to-app delegated access grants



## API Integrations

Third-party connection mapping



## Delegated Access

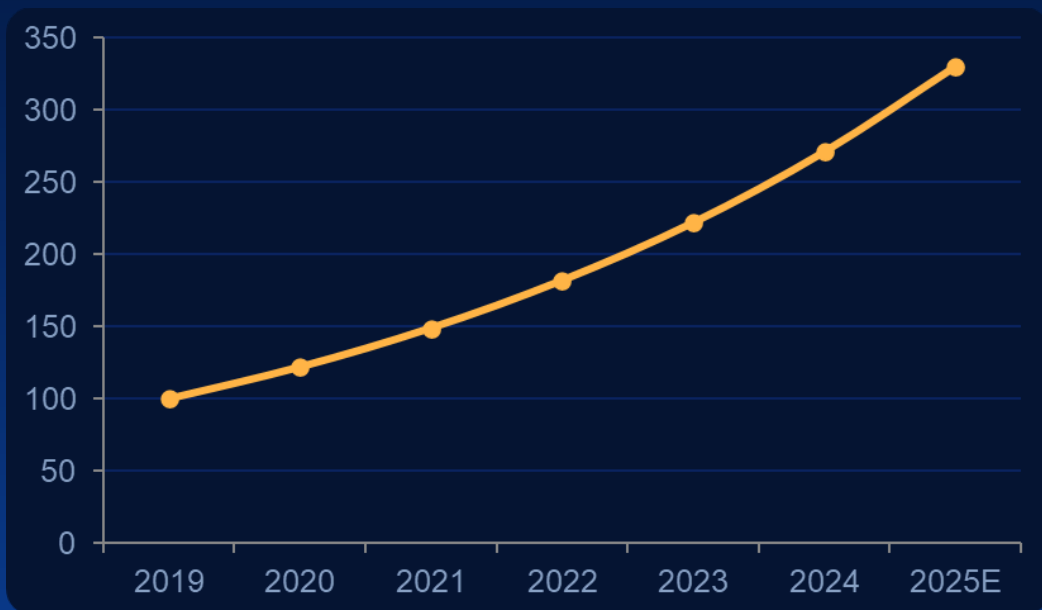
Persistent tokens & scoped permissions



## App Relationships

Hidden trust chains between SaaS apps

# SaaS Identity: Market



59%

2025 M&A Deal Volume

97%

2025 M&A Capital

22%

SaaS Cyber CAGR



SaaS Sprawl Risk

# SaaS Identity: Limitations

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Strong visibility, but limited deep enforcement capability



Dependent on provider APIs, limited to what vendors expose



Little control over underlying infrastructure credentials



Persistent OAuth tokens can bypass MFA once granted



Not end-to-end, misses behavioral and access governance layers

# SaaS Identity: Investment Activity

\$1B

2025

Proofpoint

Hornetsecurity

\$3.3B

2025

Digital Channel Security M&A

Total Category Value

Market

2025

SaaS Cyber M&A (Total)

59% of deal volume, 97% of capital deployed

# SaaS Identity: Thesis

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Strong app visibility layer, maps hidden trust chains



Grows structurally with SaaS sprawl & interconnectivity



Critical component of future unified identity platforms



Lacks deep system-level control, not a full platform alone



API dependency limits scope; insufficient without infra layer

# Access Control: Overview



Access control governs what identities are allowed to do, reducing standing privilege, minimizing blast radius, and enforcing dynamic permissioning.

## Just-In-Time Access

Temporary, scoped permissions granted on demand

## Least Privilege

Minimum access required for each task

## Entitlement Management

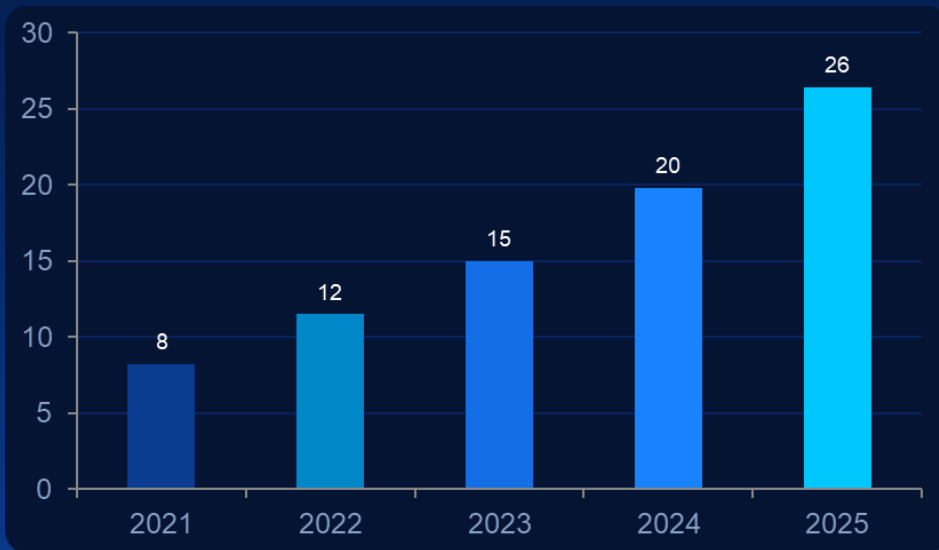
Full inventory of what each identity can access

## Dynamic Permissioning

Context-aware access based on risk signals



# Access Control: Market



\$26.4B

IAM M&A 2025

Apono

JIT Access Round

ID.me

Identity Gov Round

Durable

Budget Category

# Access Control: Limitations

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Dependent layer: requires upstream identity systems to know what exists



Does not define identity: relies on other tools for discovery & inventory



Does not monitor behavior: no visibility into post-access activity



Can create workflow friction: causing teams to bypass controls



Hard to enforce universally across cloud, on-prem, and SaaS simultaneously

# Access Control: Investment Activity

\$26.4B

2025

IAM Category

Total M&A Value: highest in category

JIT

2025

Apono

Just-in-Time Access Platform: financing round

Gov

2025

ID.me

Identity Governance: investor appetite signal

# Access Control: Thesis

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Turns identity from static authentication into dynamic permissioning



Reduces unnecessary standing access and minimizes breach blast radius



Durable compliance budget: governance is non-discretionary



Not a full platform: requires full identity visibility to work well



Best as a component of a unified identity platform, not standalone

# Behavior & Monitoring: Overview



Behavior monitoring focuses on how identities act after access is granted, detecting privilege escalation, anomalous cloud activity, and suspicious machine identity behavior.

## Privilege Escalation Detection

Flags unexpected permission changes

## Anomalous Cloud Activity

Detects unusual API calls and access patterns

## Suspicious Login Patterns

Behavioral baselines for user accounts

## NHI Behavioral Baselines

'Normal' activity profiles for machines

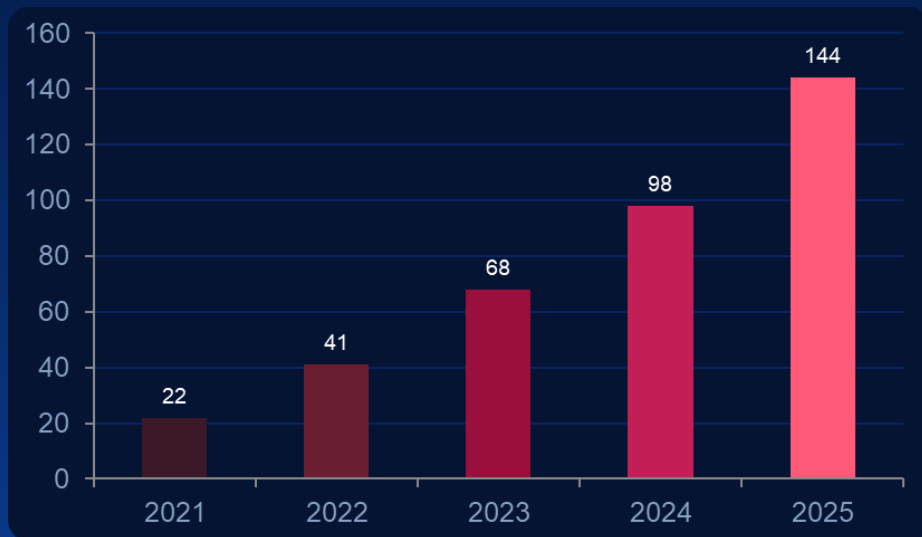
## Identity Analytics

Risk scoring across the identity surface

## ITDR (Identity Threat Detection)

Emerging response category

# Behavior & Monitoring: Market



144

AI Security Financings 2025

330+

AI Security Vendors

Top

Most Financed Cyber Sector

↑ AI

Identity Data Improves ML

# Behavior & Monitoring: Limitations

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Reactive, not preventative: detects after access is already granted



Overlaps with SIEM / UEBA / XDR, unclear standalone value



Alert fatigue risk: high false-positive rate reduces operational value



Weak NHI baselines: defining 'normal' for machines is hard



More likely to become a platform feature than a standalone winner

# Behavior & Monitoring: Investment Activity

\$700M

2025

Palo Alto Networks

Protect AI: AI/ML security & behavioral analytics

144

2025

AI Security Category

#1 financed cyber sector with 144 rounds in 2025

3 yr

Window

330+ Vendors

AI Security market emerged in under 3 years



# Behavior & Monitoring: Thesis

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Creates a data advantage, turns identity into a living risk signal



Strong fit with AI/ML security operations workflows



Valuable as a detection layer inside a broader identity platform



Primarily reactive, not a primary control layer



Stronger as a platform component than a standalone product

# Why Identity Remains Fragmented

Infrastructure Identity

SaaS Identity

Access Control

Behavior / Monitoring

← Different Control Planes →

## Different Control Planes

Cloud IAM, OAuth, PAM, and UEBA operate on separate data models with no shared identity graph.

## Built Independently

Each layer was purpose-built to solve its own problem, not designed for cross-layer integration.

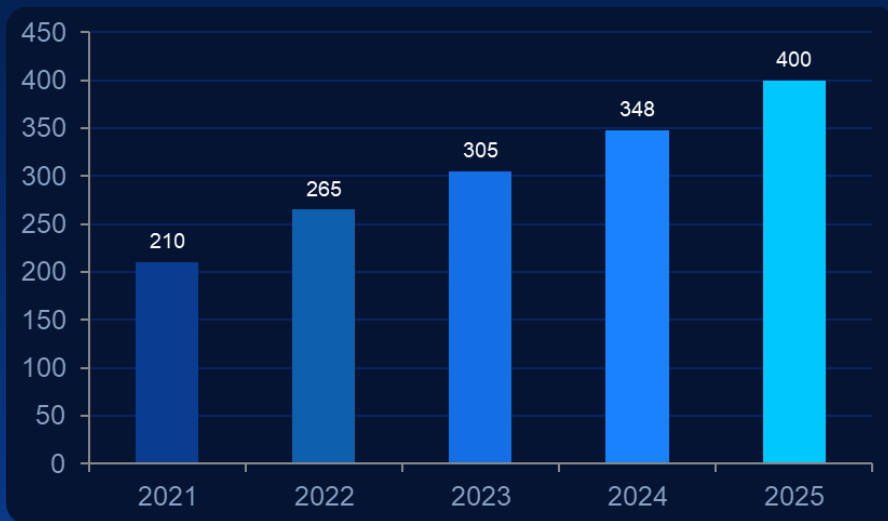
## No Unified Platform Yet

No vendor currently governs human, machine, SaaS, and AI identities in a single control plane.

## The Opportunity

Fragmentation is the market gap, the winner will unify these layers into one platform.

# Exit Landscape — 2025 Cyber M&A



400

M&A Deals (ATH)

\$96B

Disclosed Value (ATH)

92%

Strategic Buyer Capital

IPO

Platform Path to Exit



# Investment Thesis — Identity Security

1

Identity = New Control Layer —

Access, automation, and trust now flow through identities, not networks. The old perimeter is gone.

2

Market Is Fragmented —

No vendor unifies infrastructure, SaaS, access, and behavior today. The gap is the opportunity.

3

Platform Opportunity —

The highest-value company is one that expands from a strong foothold into a full identity control plane.

4

Infrastructure Is Today's Bet —

Closest to credentials and system access, the strongest starting point for platform expansion.

5

AI + NHIs Accelerate the Shift —

144:1 machine-to-human ratio and 330+ AI security vendors make this structural, not cyclical.

# Cited Sources

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1

Momentum Cyber — Cybersecurity Almanac / 2025 Year-End M&A Report

Market activity, CYBERScape taxonomy, IAM M&A value, SaaS M&A mix, AI security financings, exits, and strategic activity

2

Entro Labs — NHI & Secrets Risk Report H1 2025

144:1 NHI-to-human identity ratio and 44% YoY NHI growth data

3

Richard Stiennon / Momentum Cyber Summary

Confirmation of 59% M&A volume, 97% capital deployed, and 2025 cyber strategic activity highlights

4

Broader Market Sizing References

Cybersecurity market growth estimates and 2030 outlook sources referenced during research process